

Tyler J. Burns

tjburns@alumni.stanford.edu • Berlin, Germany

EXPERIENCE

PRESIDIO LABS Berlin, Germany
CTO and co-founder 1/23-8/23

Built out the analytical pipelines necessary for providing bioinformatics support services for academic laboratories. Developed teaching curricula to help students and postdocs learn bioinformatics.

BURNS LIFE SCIENCES CONSULTING GMBH Berlin, Germany
CEO and founder 3/20-present

Biology and bioinformatics support, research, and education. Projects have included domains of flow and mass cytometry, single-cell sequencing, high-dimensional imaging, spatial transcriptomics, multi-modal data integration, and biosecurity.

FREELANCE Berlin, Germany
Consultant 9/17-3/20

Bioinformatics and computational biology support for industry. Projects focused on high-dimensional single-cell analysis. Developed numerous data analysis pipelines accordingly.

GERMAN RHEUMATISM RESEARCH CENTER Berlin, Germany
Computational biologist 9/17-11/18

Developed analysis pipelines for mass cytometry data. Individual analysis projects included biological interpretation of results. Regularly gave institute-wide seminars and talks.

EDUCATION

Stanford University School of Medicine, Stanford, CA
PhD, Cancer Biology. Thesis laboratory: Garry Nolan 9/11-8/17

Stanford University, Stanford, CA
BA with Honors, Human Biology 9/05-12/08

SELECT PUBLICATIONS, PATENTS, SOFTWARE

My academic research focused on developing novel methods for flow and mass cytometry, both on the wet-lab and dry-lab side. This included a method for detecting nuclear localization (SLA), and a method for KNN-based analysis and visualization of mass cytometry data (Sconify).

Select Publications

- **Burns, T. J.**, Frei, A. P., Gherardini, P. F., Bava, F. A., Batchelder, J. E., Yoshiyasu, Y., et al. (2017). High-throughput precision measurement of subcellular localization in single cells. *Cytometry Part A*, 1–9. <http://doi.org/10.1002/cyto.a.23054>
- **Burns, T. J.**, Nolan G. P., Samusik N. (2018) Continuous visualization of multiple biological conditions within single cell data. *BioRxiv*.

- Budzinski L, Schulz AR, Baumgart S, **Burns T**, Rose T, Hirsland H, Mei HE (2019) Osmium-Labeled Microspheres for Bead-Based Assays in Mass Cytometry. *Journal of Immunology*. May 15;202(10):3103-3112. doi: 10.4049/jimmunol.1801640.
- Burns M, Schulz AR, Kunkel D, Hönig M, Warth S, Bengsch B, **Burns T**, Reinhardt J, Grützkau A, Yaspo ML, Sodenkamp J, Hoffmann U, Mei HE. (2020) Mass Cytometry-A Tool for the Curious: Networking in Berlin. *Cytometry Part A*. Aug;97(8):764-767. doi: 10.1002/cyto.a.24015

Patents

- Karen Sachs, Mohammed N. Al-Quaraishi, Solomon Itani, Garry P. Nolan, Sean C. Bendall, **Tyler J. Burns** Compressed Sensing for Simultaneous Measurement of Multiple Different Biological Molecule Types in a Sample. Patent number US20140106976A1.

Software

- **Burns, T. J.** (2018) Bioconductor Package “Sconify.” A toolkit for performing KNN-based statistics in flow and mass cytometry data.